# Diseases and immunity

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

## 1 · Part 1

Core · Diseases and immunity · [1]

**Answer: B**

**Explanation**

Option B matches the syllabus description: the pathogen can be passed from one host to another. Options A (deficiency disease), C (inherited disease) and D (age-related failure of body cells) involve no pathogen, so none of them is transmissible.

## 2 · Part 1

Core · Diseases and immunity · [1]

**Answer: D**

**Explanation**

Option D is indirect transmission: the pathogen reaches the worker from a contaminated surface, not from the infected person. Options A (blood entering a cut), B (kissing, body fluids) and C (body fluids during birth) are all direct contact.

## 3 · Part 1

Core · Diseases and immunity · [1]

**Answer: A**

**Explanation**

Mucus traps pathogens (which are then removed) and stomach acid kills pathogens swallowed with food or drink. Option B swaps the two roles. Options C and D wrongly say that mucus or stomach acid makes antibodies; antibodies are made by white blood cells.

## 4 · Part 1

Extended · Diseases and immunity · [1]

**Answer: C**

**Explanation**

Each pathogen has its own antigens with specific shapes, and an antibody only fits antigens whose shape is complementary to it. Option A is wrong because antibodies are made against viruses too. Option B is wrong because every pathogen has antigens. Option D is wrong because phagocytes do not destroy antibodies; the antibody simply does not fit.

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## 5 · Part 1 Extended Diseases and immunity [1]

**Answer: A**

**Explanation**

This is the syllabus description of antibodies. Option B describes phagocytes, not antibodies. Option C is wrong because antibodies are proteins, not carbohydrates. Option D is wrong because antibodies are specific: each fits only antigens of a complementary shape.

## 6 · Part 1 Extended Diseases and immunity [1]

**Answer: D**

**Explanation**

Vaccines contain weakened pathogens or their antigens, which stimulate the person's own lymphocytes to make antibodies and memory cells. Option A describes passive immunity. Options B and C are made by the person's own body, not injected in a vaccine.

## 7 · Part 1 Extended Diseases and immunity [1]

**Answer: B**

**Explanation**

In passive immunity the antibodies come from another individual, so the person's own lymphocytes do not make them, no memory cells are produced and protection is short-term. Options A and D wrongly say the person makes the antibodies, and options A and C wrongly say memory cells are produced and protection is long-term.

## 8 · Part 1 Extended Diseases and immunity [1]

**Answer: C**

**Explanation**

Active immunity is defence by antibodies the person's own body produces, gained after an infection or by vaccination. Options A, B and D all receive ready-made antibodies from another individual, which is passive immunity.

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## 9 · Part 1 Extended Diseases and immunity [1]

**Answer: D**

### Worked solution

Think of cause and effect. Nothing happens until the toxin is made. The chloride ions lower the water potential in the gut, so water moves in by osmosis. Only when large volumes of watery faeces are lost does the body become dehydrated.

## 10 · Part 1 Extended Diseases and immunity [1]

**Answer: A**

### Explanation

Cholera is caused by a bacterium and is transmitted in contaminated water. Options B and D give the wrong type of pathogen. Option C gives the wrong main route of transmission.

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## 11 · Part 1 Core Diseases and immunity [6]

**Mark scheme**

**(a)** **[1]**

- a disease in which the pathogen can be passed from one host to another (1)

**ALLOW** from one person / organism to another

**(b)** **[2]**

*Any two from:*

- contaminated surfaces (1)
- contaminated food (1)
- animals (1)
- air (1)

**ALLOW** named examples, e.g. door handles, flies, droplets from coughs or sneezes

**REJECT** water (given in the question); direct contact routes such as blood or body fluids

**(c)** **[3]**

*Any three from:*

- sewage / faeces contains pathogens (1)
- sewage treatment removes or kills pathogens before the water is returned to rivers or reused (1)
- so drinking water is not contaminated / is safe to drink (1)
- rubbish contains or encourages the growth of pathogens (1)
- rubbish attracts animals, e.g. rats or flies, that can carry pathogens to people or food (1)
- collecting rubbish removes this source / reduces contact with pathogens (1)

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**Worked solution**

(a) Use the syllabus wording: the pathogen can pass from one host to another. (b) Indirect transmission means the pathogen reaches a new host through something in between, not straight from an infected person. Apart from water, the syllabus lists contaminated surfaces, food, animals and air. (c) Damaged sewers let faeces containing pathogens get into drinking water. Treating sewage removes or kills those pathogens, so the water supply is no longer contaminated. Rotting rubbish is a place where pathogens grow, and it attracts flies and rats that carry pathogens onto people and food. Removing it takes away that source.

## 12 · Part 1 Core Diseases and immunity [5]

**Mark scheme**

**(a)** **[2]**

- mucus traps pathogens / dust (1)
- hairs (in the nose) trap / filter pathogens from the air (1)

> **ALLOW** mucus and trapped pathogens are removed from the body, e.g. by sneezing or being swallowed, as an alternative to either point

**(b)** **[1]**

- the skin is a barrier to pathogens, so a cut lets pathogens get into the body / blood (1)

**(c)** **[2]**

*Any two from:*

- white blood cells engulf / ingest pathogens (1)
- and digest / destroy them (1)

> **ALLOW** phagocytosis for both of the above points

- white blood cells produce antibodies (1)

**Worked solution**

(a) Air breathed in through the nose passes hairs and sticky mucus, and both trap pathogens before they can reach the lungs. (b) Unbroken skin stops pathogens getting in, so a cut opens a way into the tissues and blood. (c) White blood cells are the defence inside the body: some engulf and digest pathogens (phagocytosis), and others produce antibodies.

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## 13 · Part 1 Core Diseases and immunity [7]

**Mark scheme**

**(a)** **[2]**

- term 1: (48 ÷ 1200) × 100 = 4 % (1)
- term 2: (15 ÷ 1250) × 100 = 1.2 % (1)

**ALLOW** 4.0 %

**(b)** **[1]**

- the number of students eating in the canteen was different in each term (1)

**ALLOW** percentages allow a fair comparison when the totals are different

**(c)** **[4]**

- rule 1: washing hands with soap removes / kills pathogens on the hands (1)
- e.g. pathogens from faeces, so they are not transferred to the food (1)
- rule 2: raw meat may contain pathogens (1)
- keeping it separate stops pathogens transferring to cooked food, which will not be heated again to kill them (1)

**ALLOW** prevents cross-contamination for the last point

**Worked solution**

(a) Percentage = number with food poisoning ÷ number eating × 100. For term 1, 48 ÷ 1200 = 0.04, which is 4 %. For term 2, 15 ÷ 1250 = 0.012, which is 1.2 %. Check: 4 % of 1200 is 48, and 1.2 % of 1250 is 15. (b) More students ate in the canteen in term 2, so the raw numbers are not a fair comparison. Percentages take the different totals into account. (c) Hands can carry pathogens, for example from faeces after using the toilet. Washing with soap removes them so they are not passed onto food. Raw meat can carry pathogens that are killed only by cooking. If raw meat touches cooked food, those pathogens move onto food that will not be cooked again.

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## 14 · Part 1 Extended Diseases and immunity [7]

**Mark scheme**

**(a)** **[1]**

- weakened pathogens or their antigens (1)

**ALLOW** dead or inactive pathogens; parts of pathogens carrying antigens

**REJECT** antibodies

**(b)** **[2]**

- the antigens stimulate an immune response by lymphocytes (1)
- lymphocytes produce antibodies (that are specific to / complementary to the antigens) (1)

**(c)** **[2]**

*Any two from:*

- higher maximum antibody concentration after the second dose, e.g. 240 compared with 30 arbitrary units (1)
- antibody concentration increases faster after the second dose, e.g. 160 arbitrary units ten days after the second dose compared with 15 ten days after the first dose (1)
- antibody concentration stays high for longer after the second dose, e.g. still 200 arbitrary units 40 days after the second dose (1)

**ALLOW** about eight times higher maximum

**ALLOW** a maximum of 1 mark for comparisons that quote no data

**(d)** **[2]**

- memory cells were produced after the first dose (1)
- memory cells respond to the antigen quickly, so more antibodies are produced faster (1)

**ALLOW** long-term immunity provided by memory cells

**Worked solution**

(a) A vaccine contains weakened pathogens or their antigens, which cannot cause the disease. (b) The antigens stimulate lymphocytes, which respond by producing antibodies with a shape complementary to the antigens, so the antibody concentration rises. (c) Compare using numbers. After the first dose the concentration reached only 30 arbitrary units by day 20. Ten days after the second dose it was already 160, and it reached 240, which is eight times higher. It was still 200 at day 100. (d) The first dose also produced memory cells. When the same antigens enter again, the memory cells make the response faster and larger. This is why vaccination gives long-term immunity, and why a second dose is often given.

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## 15 · Part 1 Extended Diseases and immunity [6]

**Mark scheme**

**(a)** **[1]**

- as the percentage of children vaccinated increases, the number of measles cases decreases (1)

**ALLOW** negative correlation

**(b)** **[2]**

- (24 000 − 360) ÷ 24 000 × 100 (1)
- = 98.5 % (1)

**ALLOW** 99 %; 98 % only if correct working is shown

**(c)** **[3]**

*Any three from:*

- vaccinated people are immune / produce antibodies and memory cells, so they do not become infected (1)
- so they do not pass the pathogen on to others (1)
- fewer people are infected / there are fewer hosts, so the pathogen spreads less (1)
- so unvaccinated people are less likely to come into contact with the pathogen (1)

**ALLOW** herd immunity as one point, if explained

**Worked solution**

(a) Read both columns together: every time the vaccinated percentage goes up, the number of cases goes down. (b) Percentage decrease = (original − new) ÷ original × 100 = (24 000 − 360) ÷ 24 000 × 100 = 23 640 ÷ 24 000 × 100 = 98.5 %. Check backwards: 360 is 1.5 % of 24 000, and 100 − 1.5 = 98.5. (c) Vaccinated people make their own antibodies and memory cells, so they are unlikely to catch measles. If they do not catch it, they cannot pass it on. When most people are vaccinated, there are very few infected people to spread the virus, so even unvaccinated people are much less likely to meet it.

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## 16 · Part 1 Extended Diseases and immunity [6]

**Mark scheme**

**(a)** **[1]**

- across the placenta (1)

**(b)** **[3]**

*Any three from:*

- breast milk contains antibodies (1)
- the antibodies come from the mother / another individual (1)
- they protect the baby against pathogens (the mother has met) (1)
- while the baby's own immune system is still developing / before the baby can make enough of its own antibodies (1)

**(c)** **[2]**

- no memory cells are produced in passive immunity (1)
- the antibodies are broken down / not replaced, because the baby's lymphocytes did not make them (1)

**Worked solution**

(a) Before birth, antibodies cross the placenta from the mother's blood to the fetus. (b) Breast milk contains antibodies made by the mother. The baby receives them ready-made, so it is protected from pathogens straight away, while its own immune system is not yet able to make enough antibodies. (c) The baby did not make these antibodies itself, so it has no memory cells for these antigens. The antibodies are broken down over time and nothing replaces them, so the protection only lasts a short time.

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## 17 · Part 1 Extended Diseases and immunity [7]

**Mark scheme**

**(a)** **[2]**

- a bacterium (1)
- transmitted in contaminated water (1)

**REJECT** virus

**(b)** **[3]**

- the bacterium produces a toxin (1)
- the toxin causes chloride ions to be secreted into the small intestine (1)
- this lowers the water potential in the gut, so water moves into the gut by osmosis (1)

**ALLOW** water moves into the small intestine by osmosis without mention of water potential for the third point

**(c)** **[2]**

- diarrhoea causes dehydration / loss of a lot of water from the body (1)
- loss of ions from the blood (1)

**Worked solution**

(a) Cholera is caused by a bacterium and is spread in water contaminated with it. (b) In the small intestine the bacterium produces a toxin. The toxin makes the cells lining the intestine secrete chloride ions into the gut. This lowers the water potential in the gut, so water moves from the blood and tissues into the gut by osmosis, and this extra water causes diarrhoea. (c) The bacteria stay in the gut, but large amounts of water are lost in the diarrhoea, so the body becomes dehydrated. Ions are also lost from the blood. Both can be life-threatening.

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## 18 · Part 1 Extended Diseases and immunity [5]

**Mark scheme**

**(a)** **[1]**

- they have different shapes (1)

**(b)** **[2]**

- the antibodies have a shape complementary to the antigens of strain X / fit only the antigens of strain X (1)
- the antigens of strain Y have a different shape, so the antibodies cannot bind to them (1)

**(c)** **[2]**

- they cause direct destruction of the pathogen (1)
- they mark the pathogen for destruction by phagocytes (1)

**Worked solution**

(a) Each pathogen has its own antigens with specific shapes, so the two strains have antigens of different shapes. (b) The antibodies the woman made fit strain X's antigens because their shapes are complementary. Strain Y's antigens have a different shape, so those antibodies cannot bind to them, and her body has to make a new set of antibodies. (c) When antibodies bind to antigens they either destroy the pathogen directly or mark it so that phagocytes can find and destroy it.

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## 19 · Part 1 Extended Diseases and immunity [6]

**Mark scheme**

**(a)** **[2]**

- scientist 1: active immunity (1)
- scientist 2: passive immunity (1)

**(b)** **[4]**

- scientist 1 is more likely to be protected (1)
- scientist 1's own lymphocytes produced antibodies (in response to the vaccine) (1)
- memory cells were produced, giving long-term immunity / a fast response if the virus enters (1)
- scientist 2 has no memory cells / the injected antibodies are broken down and not replaced, so her protection is short-term (1)

**Worked solution**

(a) Scientist 1's body made its own antibodies after vaccination, so this is active immunity. Scientist 2 received antibodies from another person, so this is passive immunity. (b) Scientist 1 is more likely to be protected after a year. The vaccine stimulated her lymphocytes to make antibodies and memory cells, and memory cells give long-term immunity. Scientist 2's injected antibodies protect her straight away, but they are broken down over a few weeks and no memory cells were made, so her protection is short-term.

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## 20 · Part 1 Extended Diseases and immunity [7]

**Mark scheme**

**(a)** **[2]**

- district K: 60 ÷ 40 000 × 1000 = 1.5 (1)
- district L: 900 ÷ 25 000 × 1000 = 36 (1)

**(b)** **[3]**

*Any three from:*

- cholera is caused by a bacterium transmitted in contaminated water (1)
- the faeces of infected people contain the bacterium, so untreated sewage puts it into the river (1)
- people in district L drink river water containing the bacterium (1)
- in district K, water treatment / sewage treatment removes or kills the bacterium, so the drinking water is not contaminated (1)

**(c)** **[2]**

*Any two from:*

- not everyone will be vaccinated / the vaccine may not protect everyone (1)
- a clean water supply removes the source of the bacterium / stops transmission of cholera in water (1)
- a clean water supply also prevents the spread of other diseases transmitted in water (1)
- vaccination does not stop the river being contaminated by sewage (1)

**Worked solution**

(a) Cases per 1000 people = number of cases ÷ population × 1000. District K: 60 ÷ 40 000 × 1000 = 1.5. District L: 900 ÷ 25 000 × 1000 = 36. Check: 36 × 25 = 900 and 1.5 × 40 = 60. (b) Cholera bacteria leave infected people in their faeces. In district L untreated sewage flows into the river, and people drink that river water, so many swallow the bacterium. District K's water and sewage are treated, which removes or kills the bacteria, so very few people are infected. (c) A vaccination campaign will not reach everyone and may not protect everyone who is vaccinated. While the drinking water is still contaminated, the bacterium keeps reaching people. Clean water removes the source of the infection and also protects against other diseases spread in water.

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